

V2 Expansion Port Quick Reference

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Orientation used here

This reference matches the photo orientation where:

- the ****expansion port is at the top****
- the ****72-pin cartridge connector is at the bottom****

In that exact view:

- reading starts at the ****far left****
- first pair at left is `01 / 48`
- last pair at right is `24 / 25`

Pin order in this photo orientation

```
```text
haut : 01 02 03 04 05 06 07 08 09 10 11 12 13 14 15 16 17 18 19 20 21 22
 23 24
bas : 48 47 46 45 44 43 42 41 40 39 38 37 36 35 34 33 32 31 30 29 28 27
 26 25
```
```

V2 useful pins only

```
```text
haut : [01][02][03][04][05][06][07][08][09][10][11][12][13][14][15][16][17][18][19][20][21][22]
 [23][24]

 |
 A15

 |
 /IRQ

bas : [48][47][46][45][44][43][42][41][40][39][38][37][36][35][34][33][32][31][30][29][28][27]
 [26][25]

 | | | | |
 | | 5V GND 02 01 00

 D6 D7

 D0 D1 D2 D3 D4 D5
```
```

Recommended proto 1 safe bundle

This is now the ****recommended V2 proto 1 safe bundle****.

The idea is:

- keep the active protocol simple
- use `OUT0..OUT2` as command
- use `/OE2` plus the `PORT1-0..4` group as the read side for `\$4017`
- keep only a ****small sniff bundle****: `CPU D0`, `CPU D1`, `CPU D2`
- postpone `/IRQ` to a later step

```
```text
haut : [01][02][03][04][05][06][07][08][09][10][11][12][13][14][15][16][17][18][19][20][21][22]
 [23][24]

 | | |
 GND A15 OE2

 | | |
 P1-2 P1-3 P1-4

bas : [48][47][46][45][44][43][42][41][40][39][38][37][36][35][34][33][32][31][30][29][28][27]
 [26][25]

 | | | | |
 5V GND 02 01 00

 CD0 CD1 CD2
```
```

Additional lines on the same safe proto bundle:

```
```text
`19 = PORT1-0`
`20 = PORT1-1`
```
```

So the ****safe first-choice wiring set**** is:

```
1. `48` = `+5V`
2. `47` = `GND`
3. `02` = `GND` secondaire
4. `45` = `OUT2`
5. `44` = `OUT1`
6. `43` = `OUT0`
7. `11` = `/OE2` (`$4017` read strobe)
8. `19` = `PORT1-0`
9. `20` = `PORT1-1`
10. `15` = `PORT1-2`
11. `16` = `PORT1-3`
12. `18` = `PORT1-4`
13. `05` = `A15` bonus debug
14. `32` = `CPU D0` sniff
15. `31` = `CPU D1` sniff
16. `30` = `CPU D2` sniff
```

What is already soldered in the current proto

```
- `48` = `+5V`
- `47` = `GND`
- `45` = `OUT2`
- `44` = `OUT1`
- `43` = `OUT0`
- `19` = `PORT1-0`
- `20` = `PORT1-1`
- `15` = `PORT1-2`
- `16` = `PORT1-3`
- `18` = `PORT1-4`
```

What remains to solder now

```
1. `02` = `GND` secondaire
2. `11` = `/OE2`
3. `05` = `A15`
4. `32` = `CPU D0` sniff
5. `31` = `CPU D1` sniff
6. `30` = `CPU D2` sniff
```

Current color-coded ribbon order

The correct mental model is:

- this is a ****16-wire ordered bundle****
- colors form a repeated sequence
- the safe way to read it is ****by ribbon order****, not "one color = one unique signal"

Ribbon order currently understood from the latest diagram:

```
1. violet = `02` = `GND2`
2. grey = `30` = `CPU D2`
3. white = `31` = `CPU D1`
4. black = `32` = `CPU D0`
5. brown = `11` = `/OE2`
6. red = `05` = `A15`
```

```
7. orange = `18` = `Joypad D4`
8. yellow = `16` = `Joypad D3`
9. green = `15` = `Joypad D2`
10. blue = `20` = `Joypad D1`
11. violet = `19` = `Joypad D0`
12. grey = `43` = `OUT0`
13. white = `44` = `OUT1`
14. black = `45` = `OUT2`
15. brown = `47` = `GND`
16. red = `48` = `+5V`
```

Important:

- `Joypad D0..D4` here corresponds to the five data lines already soldered:
 - `19`, `20`, `15`, `16`, `18`
- the repeated colors are normal in this scheme
- what matters is the ****position in the ribbon****

Advanced bundle (more powerful, more risky)

This is the other direction we discussed, but ****not**** the recommended bring-up path:

- `CPU D0..D7`
- `A15`
- `/IRQ`
- later `R/W` and `M2`

Keep this for a later bus-aware proto, not for proto 1.

Signal list

Top row

- `02` = `GND`
- `05` = `A15`
- `11` = `/OE2` (`\$4017` read strobe)
- `15` = `PORT1-2`
- `16` = `PORT1-3`
- `18` = `PORT1-4`
- `19` = `PORT1-0`
- `20` = `PORT1-1`

Bottom row

- `48` = `+5V`
- `47` = `GND`
- `45` = `OUT2`
- `44` = `OUT1`
- `43` = `OUT0`
- `32` = `CPU D0`
- `31` = `CPU D1`
- `30` = `CPU D2`
- `29` = `CPU D3`
- `28` = `CPU D4`
- `27` = `CPU D5`
- `26` = `CPU D6`
- `25` = `CPU D7`

Quick reality check before soldering all lines

- `02` and `47` should both be `GND`
- `43`, `44`, `45` should be the three contiguous `OUT0`, `OUT1`, `OUT2` lines
- in this ****proto 1 safe**** version, `11` is the important read strobe line for `\$4017`
- `19`, `20`, `15`, `16`, `18` are the five controller data lines we are actively using here
- `/IRQ` is intentionally ****not**** part of this first `16-wire` bundle

If that matches your meter, the orientation is locked.

Cable / connector note

For a first V2 bundle, we need roughly `16` lines.

What exists in standard parts:

- `16-conductor ribbon cable`
- `HE10 / IDC 16-pin connectors`
- `16-pin IDC board headers`

What probably does ****not**** exist as a neat off-the-shelf "perfect V2 cable":

- a ready-made round cable already terminated exactly for this project

So the realistic options are:

1. flat `16-way ribbon` with IDC/HE10 ends
2. ribbon cable that is separated for a short distance near the motherboard
3. custom round multicore cable only if we later want a cleaner final build

Sources

- https://www.nesdev.org/wiki/Expansion_port
- <https://commons.wikimedia.org/wiki/File:Nintendo-NES-Mk1-Motherboard-Bottom.jpg>
- <https://www.e44.com/connectique/connecteurs/fiches/IDC-16b/Femelle/he10-femelle-sertir-16-pin-s-HE10F16.html>
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